

Note on MLS eqn. (2.18) p. 30

MLS state that eqn. (2.18) can be used if $\theta \neq 0$. They should really have stated that it cannot be used if $\sin \theta = s_\theta \neq 0$. One can distinguish the following 3 cases where one wishes to calculate the column vector ω , after having solved for θ with equation (2.17):

- when $\sin \theta = s_\theta \neq 0$; in this case eqn. (2.18) can be used.
- when $\sin \theta = 0$ because $\theta = 0$: in this case there is no rotation and R is a 3×3 identity matrix. In this case ω does not have any meaning and, for purposes of completing the calculation, can be assigned an arbitrary unit vector.
- when $\sin \theta = 0$ because $\theta = \pi$ rad: in this case $c_\theta = \cos \theta = -1$ and $v_\theta = 2$, which means that ω can be solved by setting the diagonal elements of the last matrix in eqn. (2.16) equal to the corresponding elements of the matrix R , leading to:

$$\omega_1 = \sqrt{\frac{r_{11} - c_\theta}{v_\theta}} \quad \omega_2 = \sqrt{\frac{r_{22} - c_\theta}{v_\theta}} \quad \omega_3 = \sqrt{\frac{r_{33} - c_\theta}{v_\theta}}$$